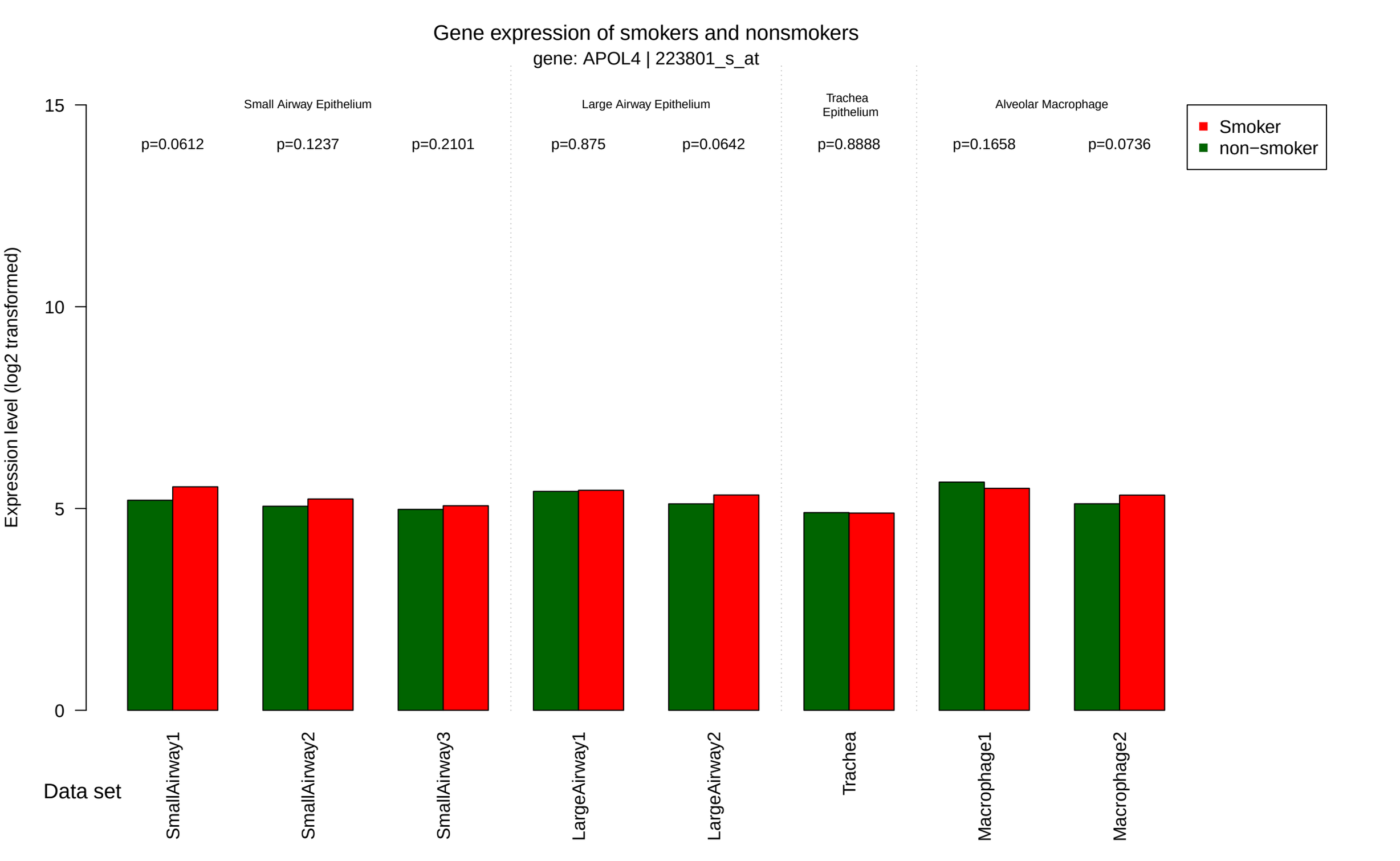
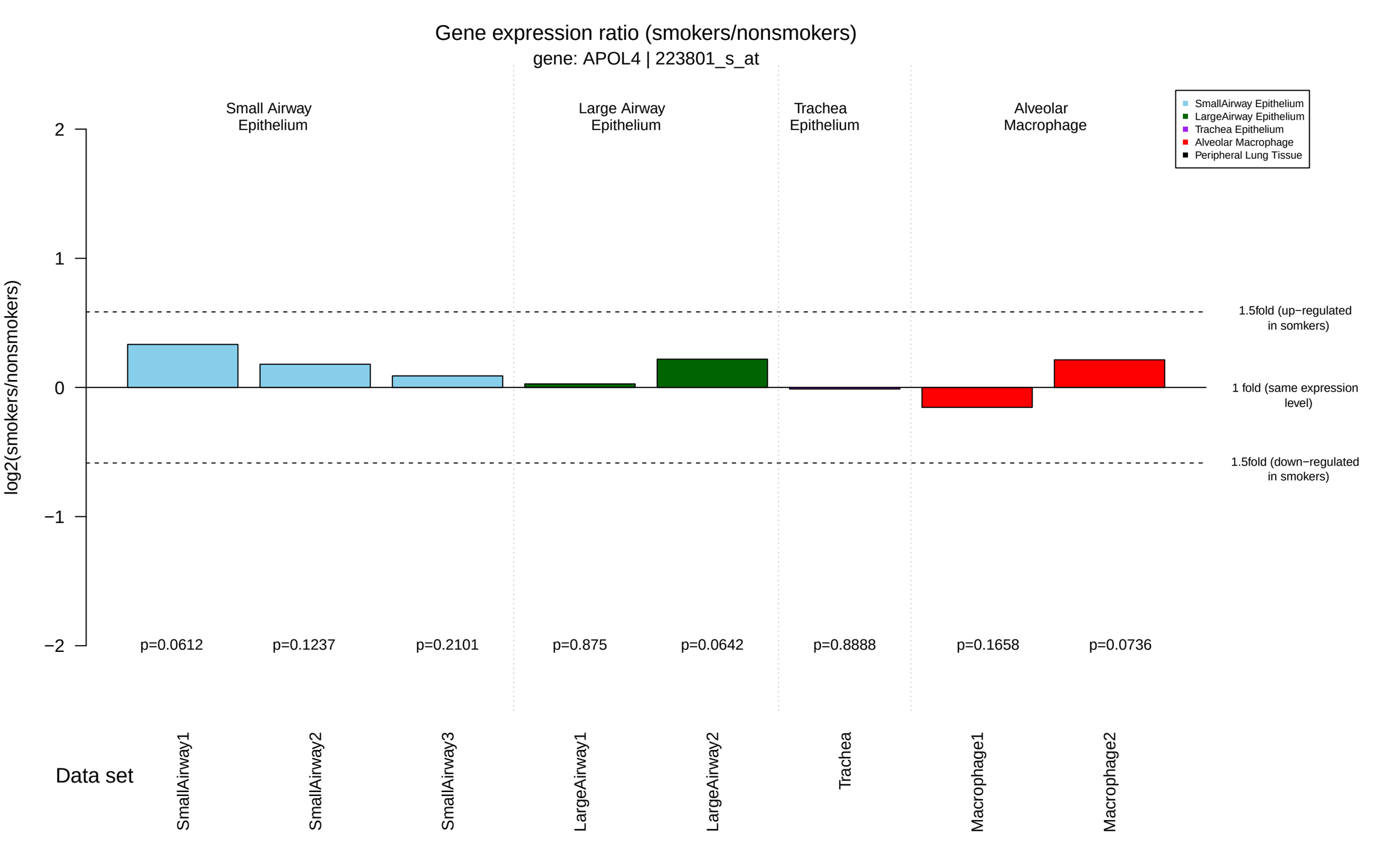
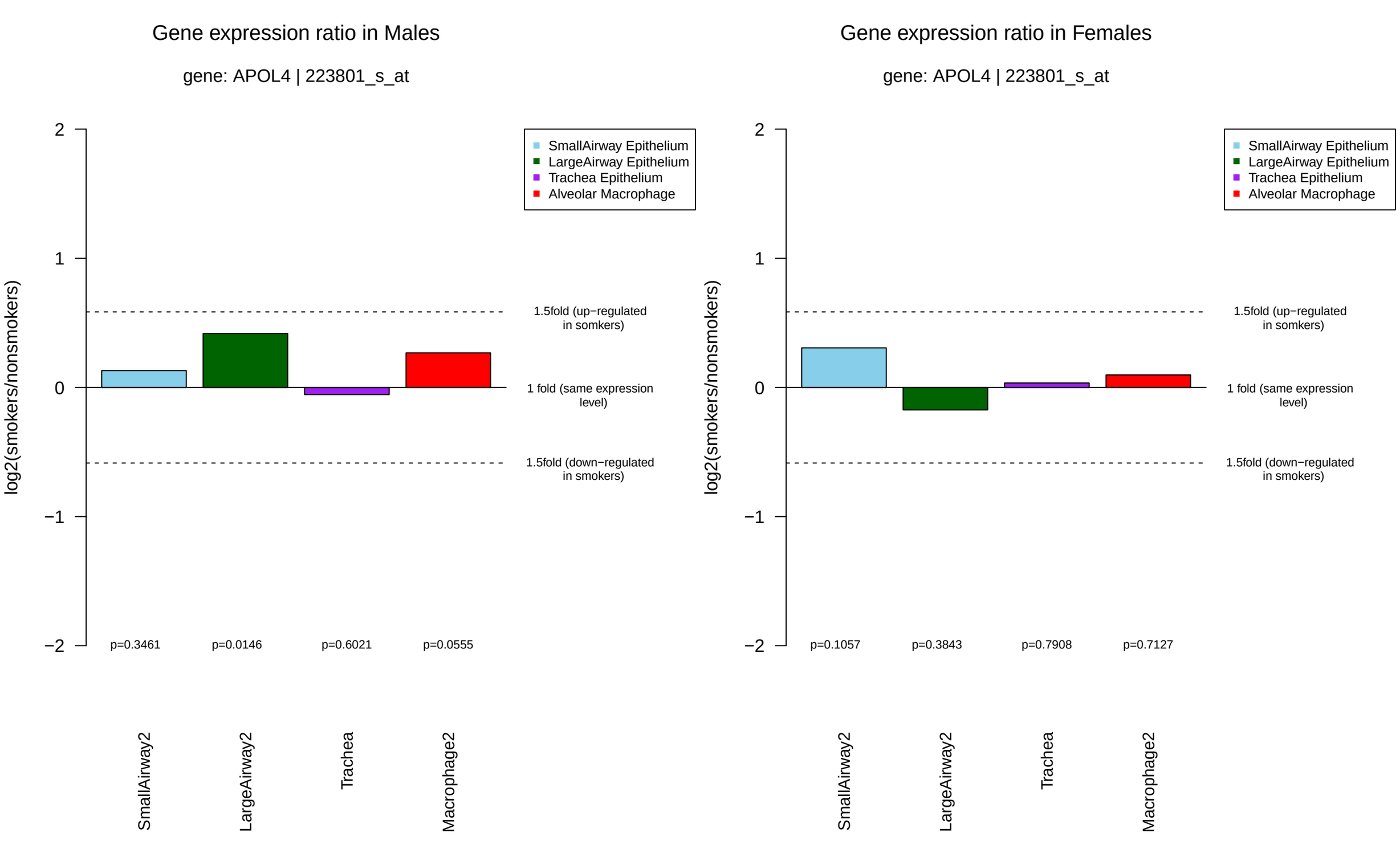
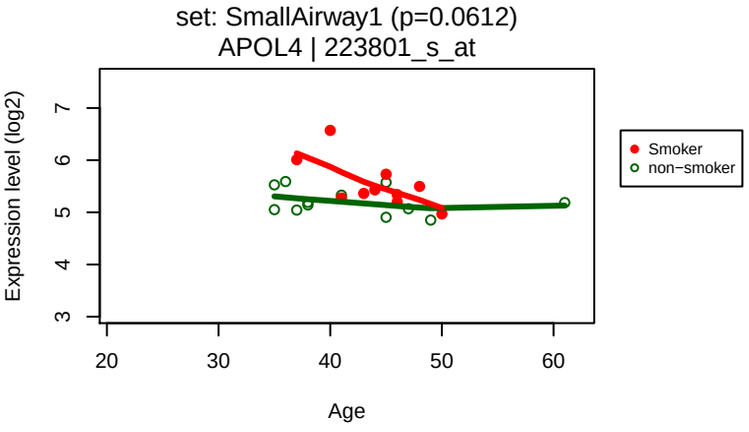
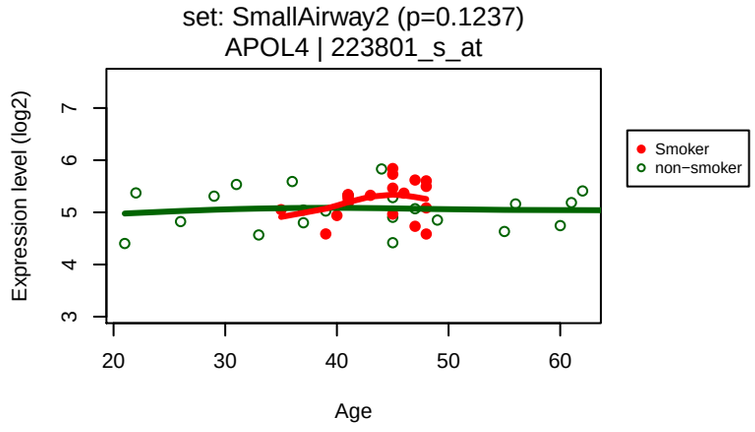
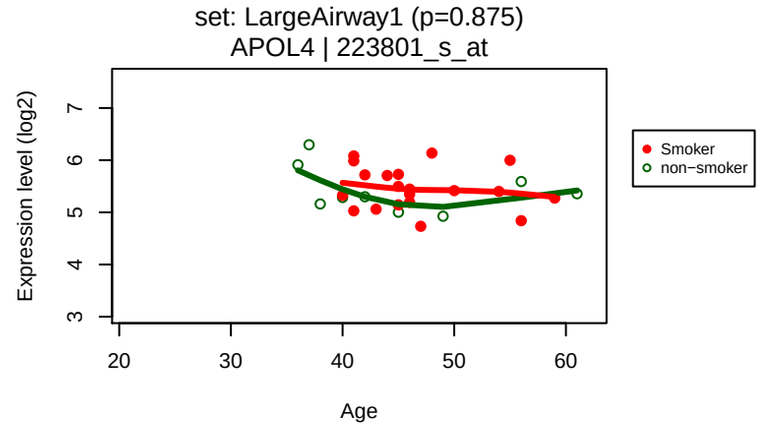

Gene expression of smokers and nonsmokers
gene: APOL4 | 223801_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0612
p=0.1237
p=0.2101
p=0.875
p=0.0642
p=0.8888
p=0.1658
p=0.0736
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

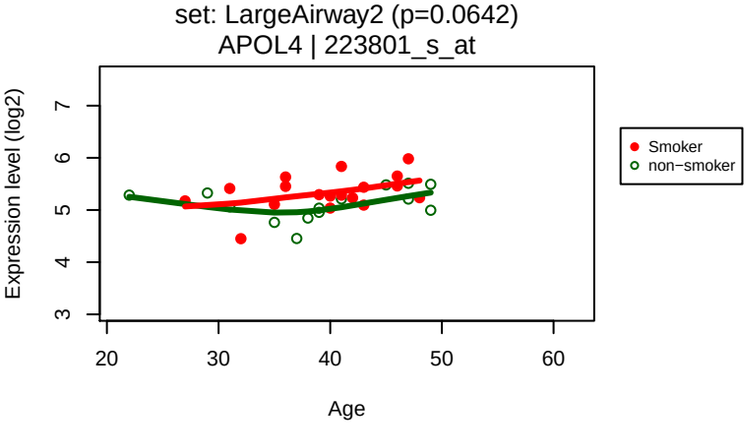

Gene expression ratio (smokers/nonsmokers)
gene: APOL4 | 223801_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0612
p=0.1237
p=0.2101
p=0.875
p=0.0642
p=0.8888
p=0.1658
p=0.0736
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

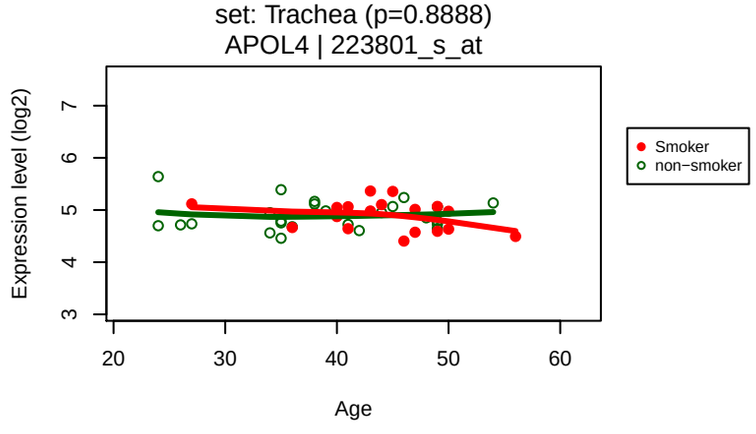

Gene expression ratio in Males
gene: APOL4 | 223801_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3461
p=0.0146
p=0.6021
p=0.0555
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APOL4 | 223801_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1057
p=0.3843
p=0.7908
p=0.7127
SmallAirway2
LargeAirway2
Trachea
Macrophage2

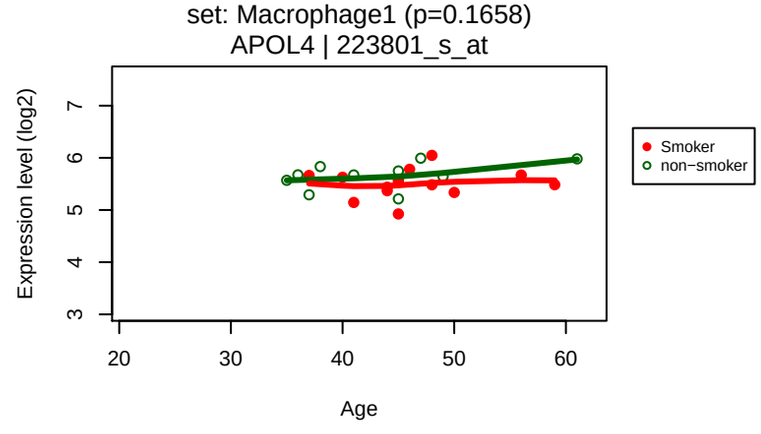

set: SmallAirway1 (p=0.0612)
APOL4 | 223801_s_at
Expression level (log2)
Age
Smoker
non−smoker

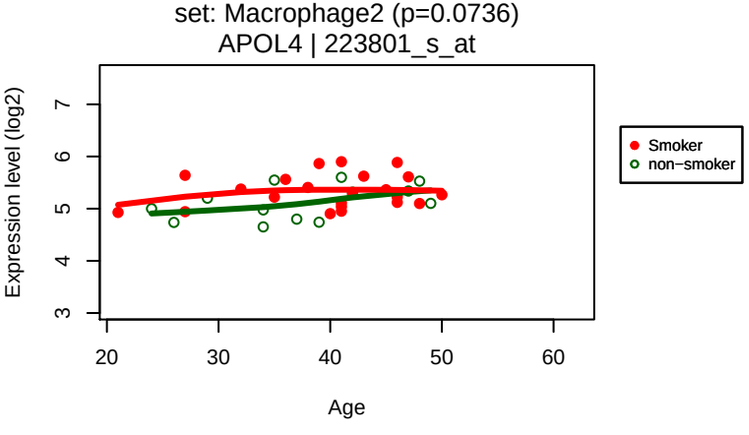

set: SmallAirway2 (p=0.1237)
APOL4 | 223801_s_at
Expression level (log2)
Age
Smoker
non−smoker

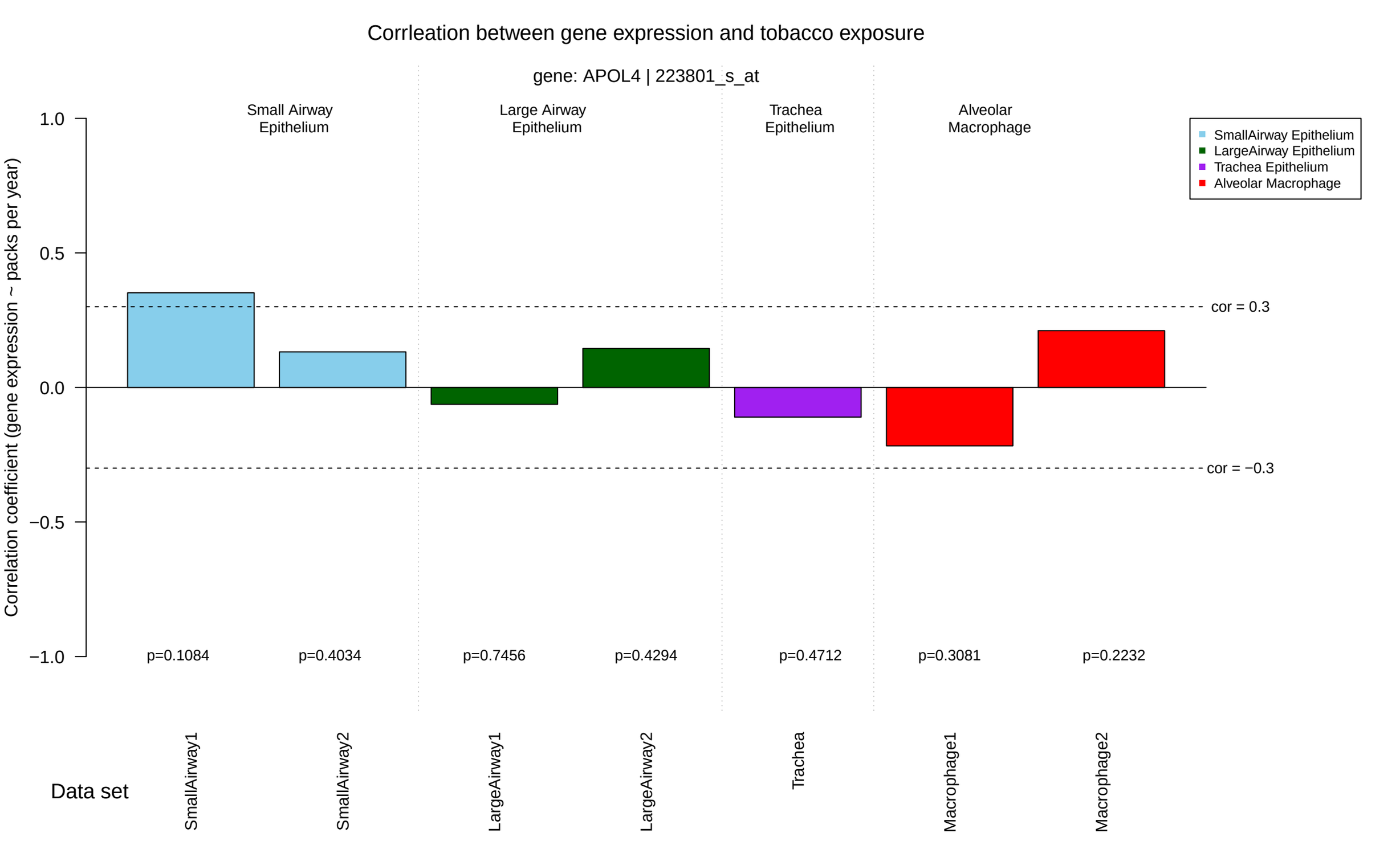

set: LargeAirway1 (p=0.875)
APOL4 | 223801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0642)
APOL4 | 223801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8888)
APOL4 | 223801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1658)
APOL4 | 223801_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0736)
APOL4 | 223801_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APOL4 | 223801_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1084
p=0.4034
p=0.7456
p=0.4294
p=0.4712
p=0.3081
p=0.2232
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2